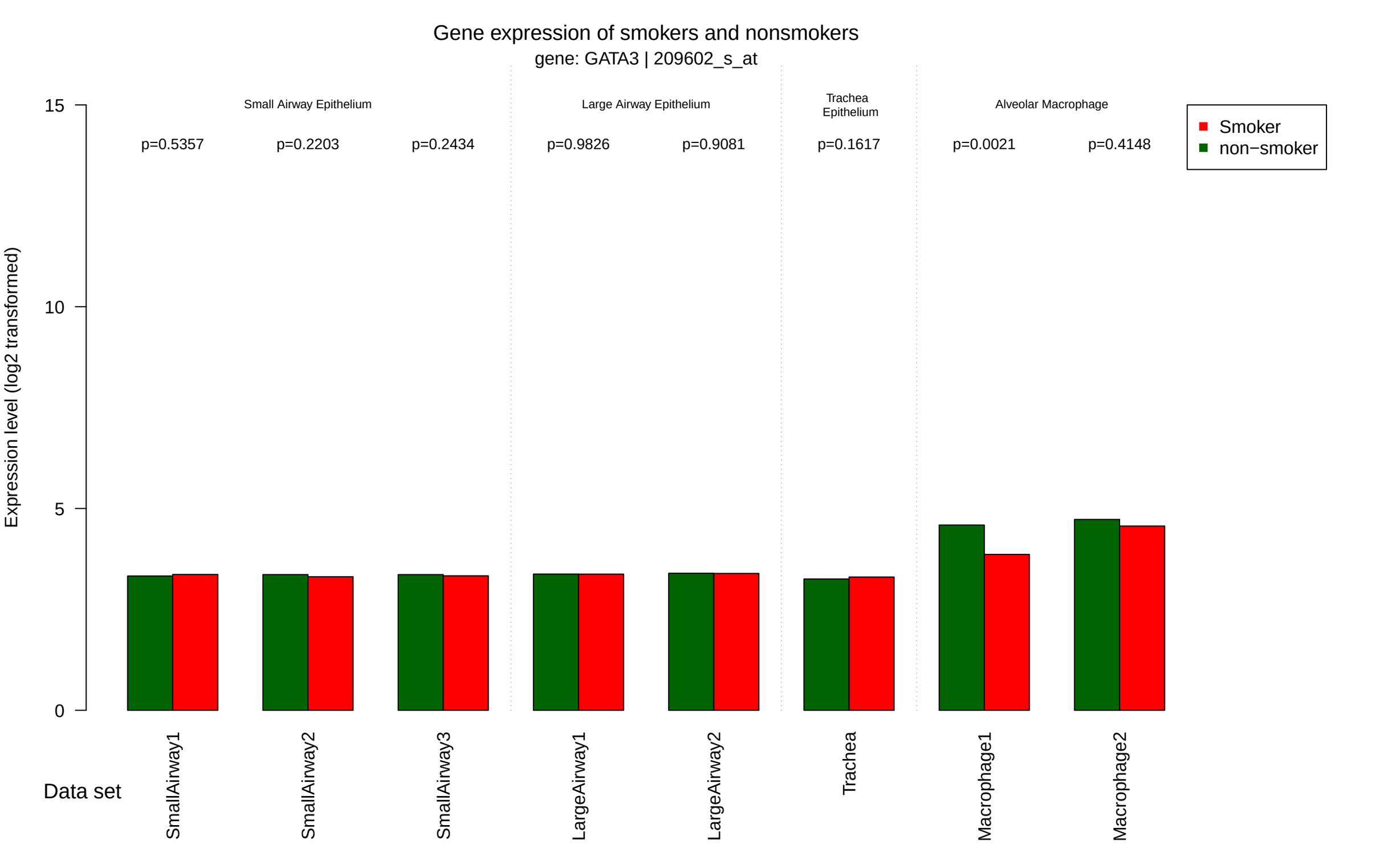
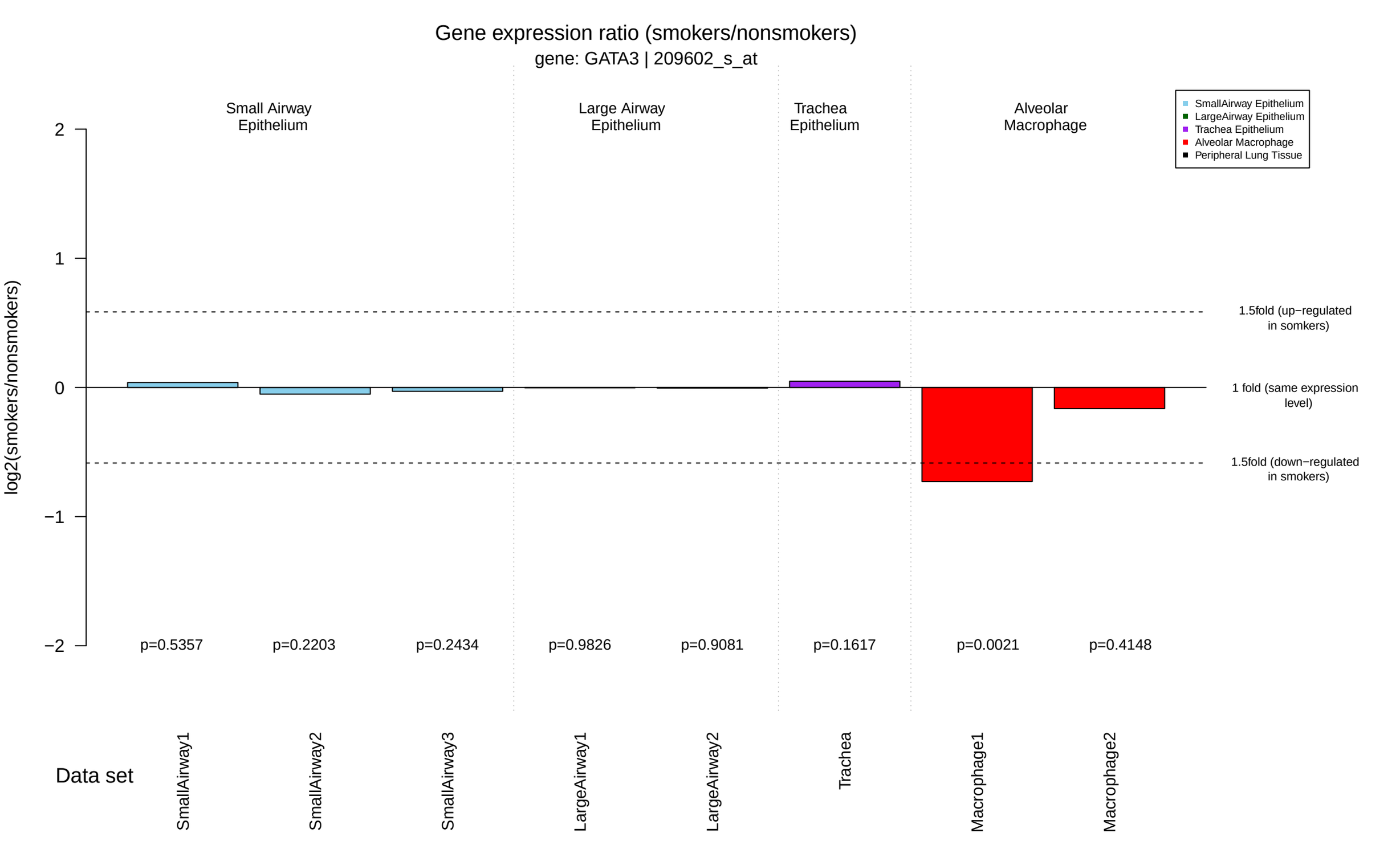
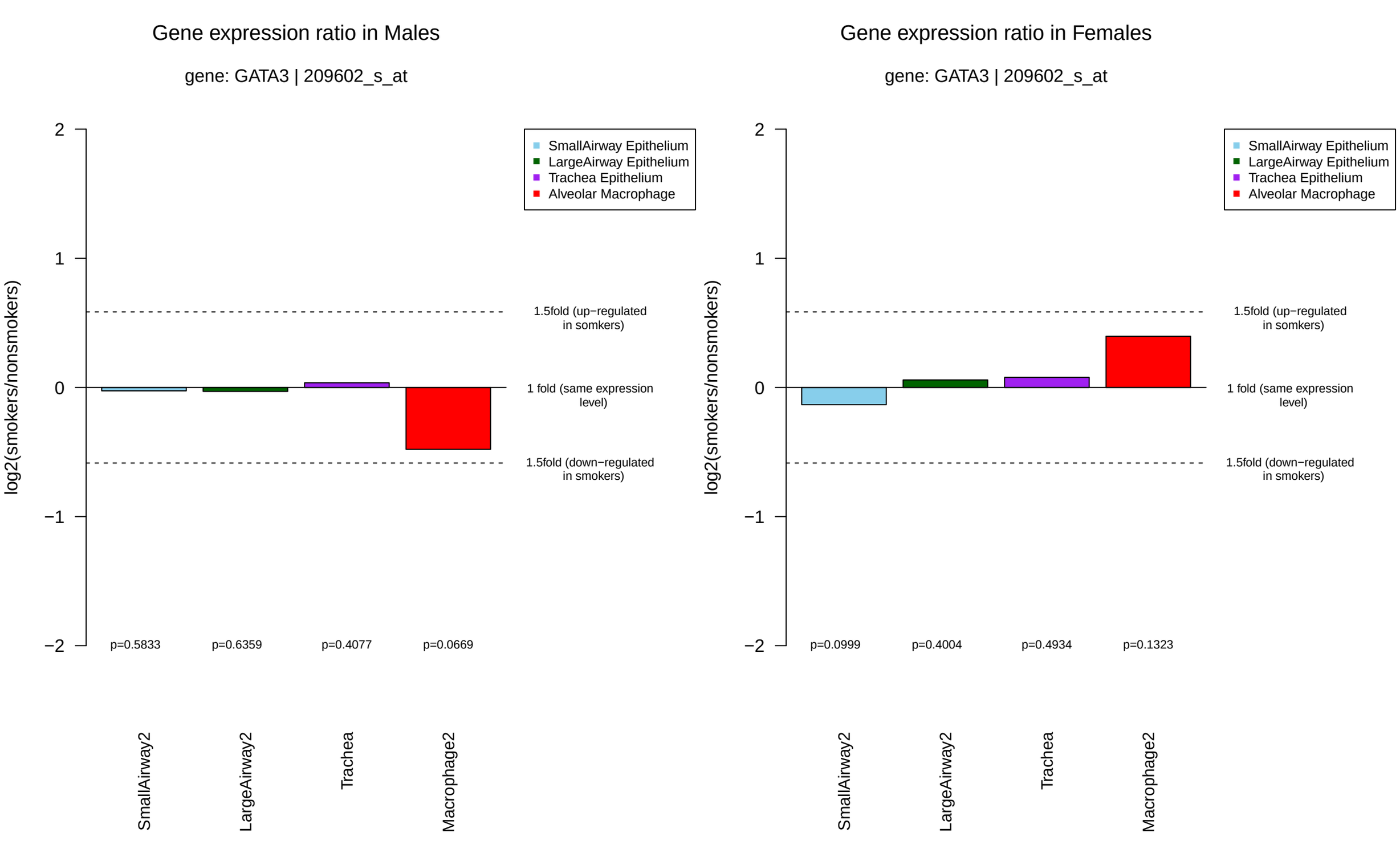
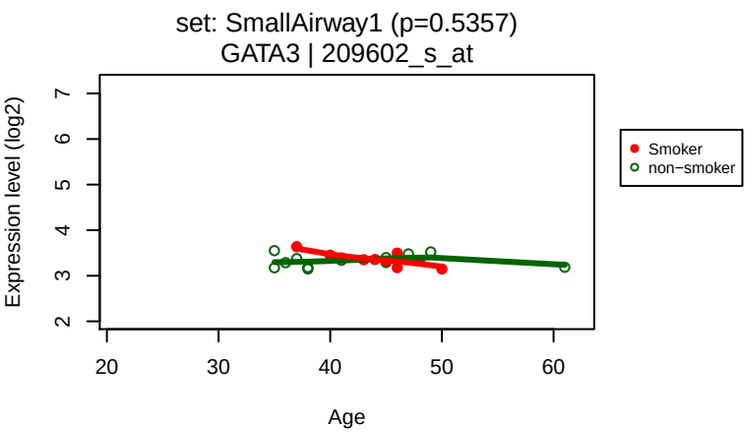
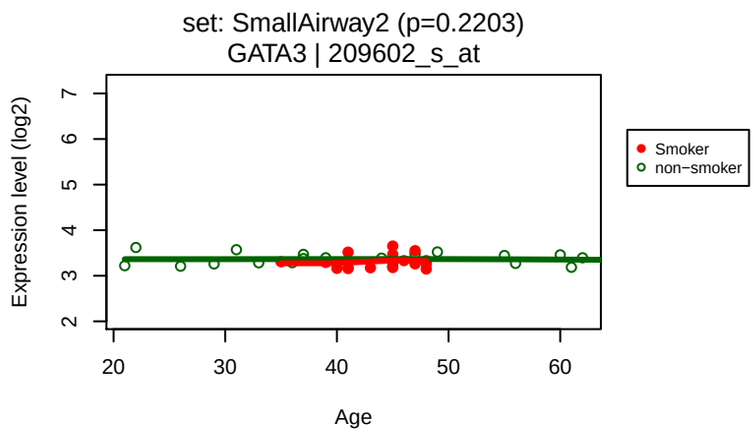
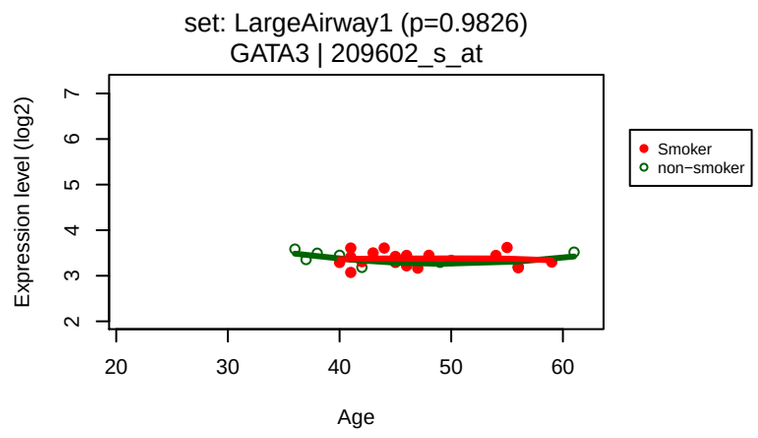

Gene expression of smokers and nonsmokers
gene: GATA3 | 209602_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5357
p=0.2203
p=0.2434
p=0.9826
p=0.9081
p=0.1617
p=0.0021
p=0.4148
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

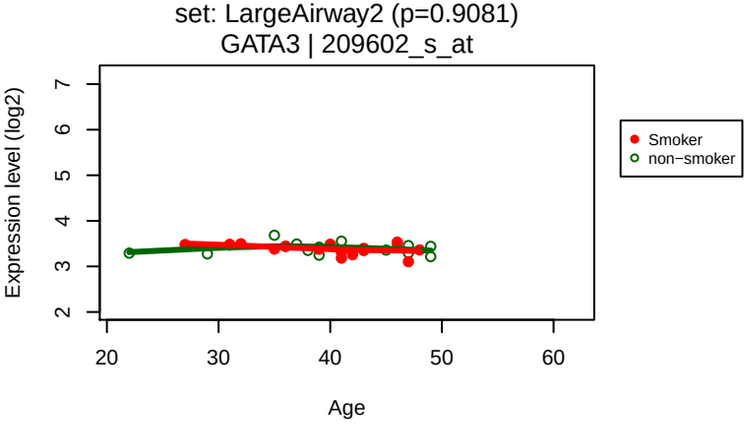

Gene expression ratio (smokers/nonsmokers)
gene: GATA3 | 209602_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5357
p=0.2203
p=0.2434
p=0.9826
p=0.9081
p=0.1617
p=0.0021
p=0.4148
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

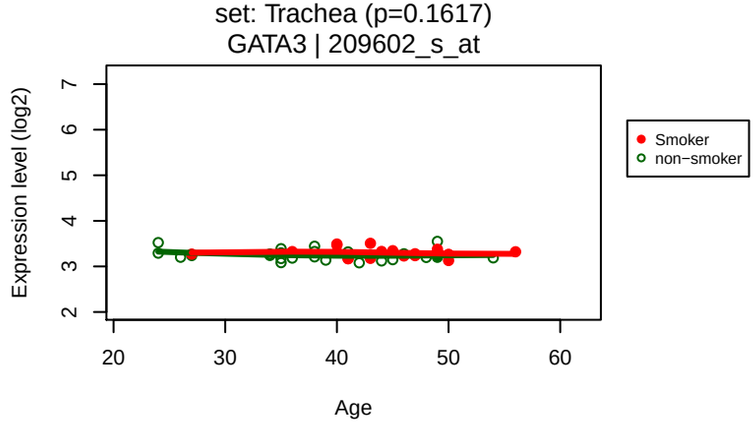

Gene expression ratio in Males
gene: GATA3 | 209602_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5833
p=0.6359
p=0.4077
p=0.0669
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GATA3 | 209602_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0999
p=0.4004
p=0.4934
p=0.1323
SmallAirway2
LargeAirway2
Trachea
Macrophage2

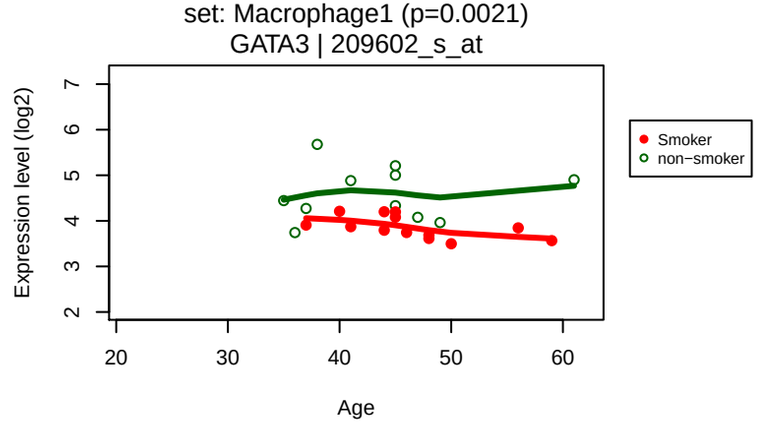

set: SmallAirway1 (p=0.5357)
GATA3 | 209602_s_at
Expression level (log2)
Age
Smoker
non−smoker

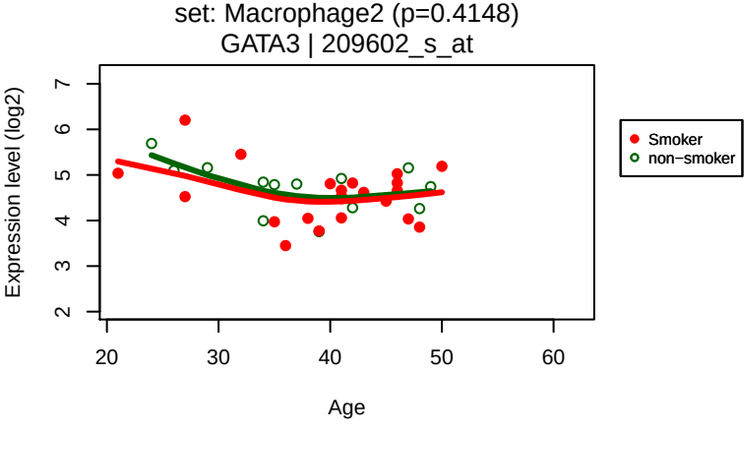

set: SmallAirway2 (p=0.2203)
GATA3 | 209602_s_at
Expression level (log2)
Age
Smoker
non−smoker

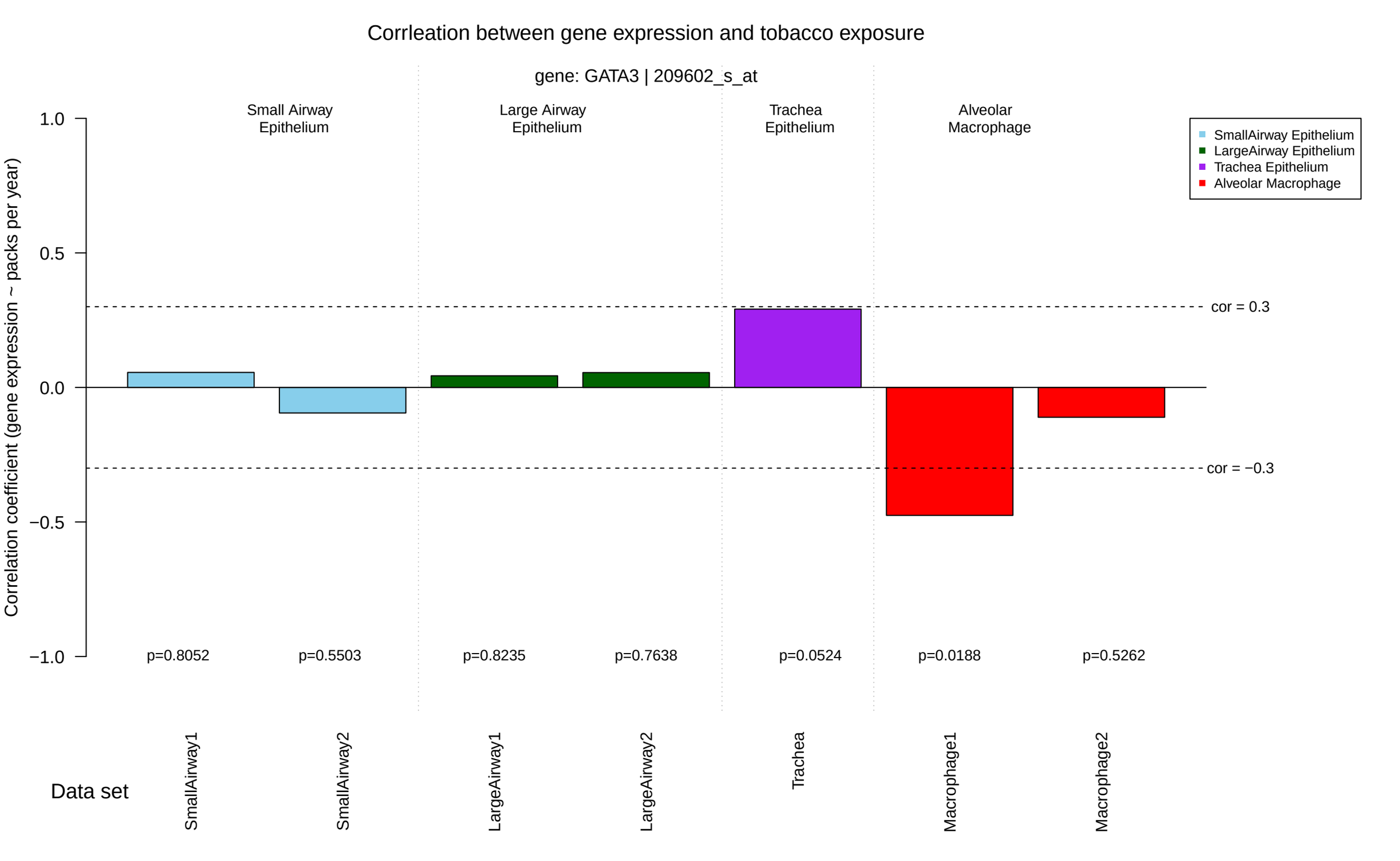

set: LargeAirway1 (p=0.9826)
GATA3 | 209602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9081)
GATA3 | 209602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1617)
GATA3 | 209602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0021)
GATA3 | 209602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4148)
GATA3 | 209602_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GATA3 | 209602_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8052
p=0.5503
p=0.8235
p=0.7638
p=0.0524
p=0.0188
p=0.5262
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2